

M&S: Final Assignment

Group 05

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1 Introduction

This report contains the modelling and simulation of a 3 DOF robot arm modelled in 20-sim. The model is constructed step by step, after each addition, the model is verified to ensure proper workflow and allow for easy debugging. The focus in this particular assignment is to make the robot arm as realistic as possible for its intended use, the drawing of a circle.

The focus of the modelling is on the dynamics of the system, including the discrete controller behaviour. The tuning of the PD controller is not performed thoroughly, since it is not the goal of this paper. Furthermore, the thermal domain is neglected.

2 Model

This section provides a generic overview of the model, including modelling decisions, and their justification. The section starts with a general model overview, showing the 20-sim model. After that, more information is provided on the robot manipulator model. Next, the PD controller and torque calculation are shown, followed by the gravity compensation and the SEA models. Lastly, the model discretization is discussed.

2.1 General overview

A general overview of the model is given in Figure 1. For clarity, the model is split up in multiple parts, including the reference trajectory, the PD controller, a torque calculation, the SEA models and the robot manipulator model. Lastly, the model also contains the `GlobalsSEA_qx`, with x a number between 1 and 3, one for each SEA model.

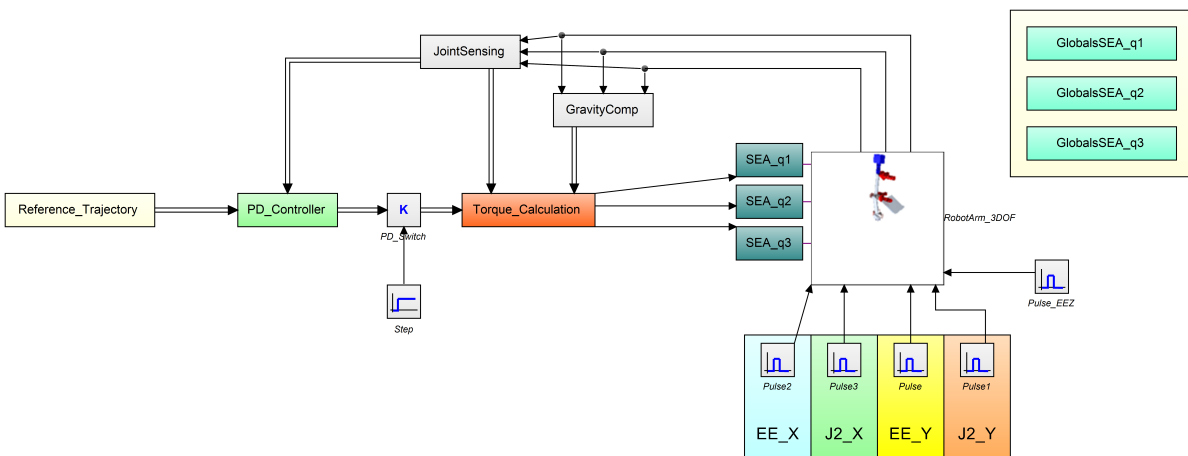


Figure 1: The full 20-sim model, separated into sub-models.

As stated before, the following sub-sections will dive deeper into all these sub-models. Before that, consider the `GlobalsSEA_qx` blocks. These were implemented to keep all SEA variables neatly organized, allowing for a quick change of parameters in an organized way. These sub-models are nothing more than a way of storing these global parameters for easy adjustment. Furthermore, four impulse sources are also present, allowing for force impulses on the robot end effector and second joint. More on that in subsubsection 2.2.3.

2.2 Robot manipulator

2.2.1 System description

Before getting into the theory, and the modelling of it, a system description is presented. The system consists of three joints, with 3 rigid bodies. The system itself is shown in Figure 2. In zero configuration, where $q_1 = q_2 = q_3 = 0$, so all joints are at zero angle, the robot arm is pointing upwards.

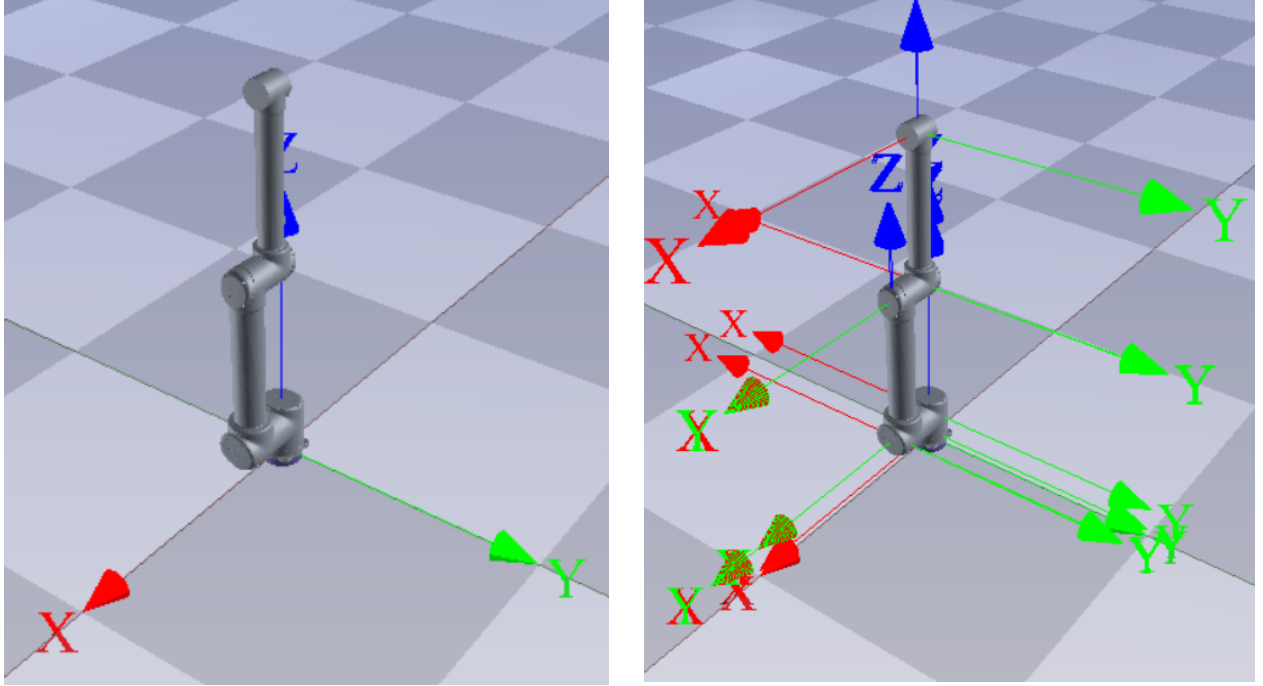


Figure 2: The robot arm without and with the internal reference frames. The rotation axes are parallel with global z, x, and x respectively, which will be denoted as ZXX.

The parameters of the robot arm links are based on the UR10 made by Universal Robots [1]. These parameters are shown in Table 1.

Table 1: Mass, lengths and inertias of the robot arm links.

	link 1 (base)	link 2 (shoulder)	link 3 (elbow)
mass m [kg]	4.00	12.7	4.27
link length L [m]	0.038	0.612	0.5723
CoM location l_c [m]	0.021	0.232	0.3323
I_{xx} [kg m ²]	0.03408	0.02814	0.01014
I_{yy} [kg m ²]	0.03529	0.77068	0.30928
I_{zz} [kg m ²]	0.02156	0.76943	0.30646
I_{xy} [kg m ²]	0.00002	0.00005	0.00008
I_{xz} [kg m ²]	-0.00425	-0.01561	0.00916
I_{yz} [kg m ²]	0.00008	0.00002	0.00000

2.2.2 Theoretical background

Modelling the robot arm itself requires little theoretical background, due to the usage of the 3D-mechanics toolbox. However, for later derivation of gravity compensation and torque control, a geometric Jacobian is required. Consider the point where $q_1 = q_2 = q_3 = 0$, so the zero configuration, as the robot arm standing

completely upright in global z . Deriving the geometric Jacobian requires the unit joint twists in the world fixed frame. The unit twists can be calculated by using the relation of a one dimensional joint to the twist given by

$$\hat{T}_i^{0,(i-1)} = \begin{pmatrix} \hat{\omega} \\ r \wedge \hat{\omega} \end{pmatrix} \in \mathbb{R}^6 \quad (1)$$

where $\hat{\omega}$ is the unit rotation axis of the joint, and r is the distance to that axis from the origin of the base frame. For this specific model, the geometric Jacobian can be expressed as

$$J(\mathbf{q}) = \begin{bmatrix} \hat{T}_1^{0,0} & \hat{T}_2^{0,1} & \hat{T}_3^{0,2} \end{bmatrix} \in \mathbb{R}^{6 \times 3} \quad (2)$$

For completeness, the full Jacobian of this specific system will be derived now, which is later used for both torque calculations and gravity compensation. The first and second unit twists can be derived easily, resulting in

$$\hat{T}_1^{0,0} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \hat{T}_1^{0,2} = \begin{pmatrix} \cos(q_1) \\ \sin(q_1) \\ 0 \\ -l_1 \cos(q_1) \\ l_1 \sin(q_1) \end{pmatrix} \quad (3)$$

For the third unit twist $\hat{T}_2^{0,3}$, getting r is a little more involved. Deriving the position of the rotation axis yields in the fixed frame, expressed in the fixed frame, gives

$$r_0^{0,3} = \begin{pmatrix} l_2 \sin(q_2) \sin(q_1) \\ l_2 \cos(q_2) \cos(q_1) \\ l_1 + l_2 \cos(q_2) \end{pmatrix} \quad (4)$$

Using the known relation for the unit twist, and the fact that the rotation axis is parallel to the y -axis at $q_1 = 0$, the following result is obtained for both the unit twist, and the geometric Jacobian

$$\hat{T}_3^{0,2} = \begin{pmatrix} \cos(q_1) \\ \sin(q_1) \\ 0 \\ -\sin(q_1)(l_1 + l_2 \cos(q_2)) \\ \cos(q_1)(l_1 + l_2 \cos(q_2)) \\ 0 \end{pmatrix} \quad (5)$$

$$J(\mathbf{q}) = \begin{bmatrix} 0 & \cos(q_1) & \cos(q_1) \\ 0 & \sin(q_1) & \sin(q_1) \\ 1 & 0 & 0 \\ 0 & -l_1 \cos(q_1) & -\sin(q_1)(l_1 + l_2 \cos(q_2)) \\ 0 & l_1 \sin(q_1) & \cos(q_1)(l_1 + l_2 \cos(q_2)) \\ 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{6 \times 3} \quad (6)$$

To obtain the Jacobian with respect to the end effector, a simple transformation can be used. The Jacobian can be transformed to the end effector position p_{EE} by using the following transformation

$$J_{EE}(\mathbf{q}) = \begin{bmatrix} I_{3 \times 3} & 0_{3 \times 3} \\ -[\mathbf{r}_{0,EE} \times] & I_{3 \times 3} \end{bmatrix} J(\mathbf{q}) \quad (7)$$

where $r_{0,EE}$ is the distance to the end effector position expressed in the fixed frame, from the origin of the fixed frame, and the $\times$ operator is used to indicate the skew symmetric matrix form of this vector. This vector is obtained numerically, so a derivation will not be shown. However, in terms of indices, due to the structure of this transformation, it can be rewritten into the form

$$J_{EE}(\mathbf{q})[1:3, 1:3] = J(\mathbf{q})[1:3, 1:3] \quad (8)$$

$$J_{EE}(\mathbf{q})[4:6, 1:3] = J(\mathbf{q})[4:6, 1:3] - [\mathbf{r}_{0,EE} \times] J(\mathbf{q})[1:3, 1:3] \quad (9)$$

where the square brackets indicate said indices.

2.2.3 Modelling choices

The joints were chosen in ZXX (q_1 parallel to the z axis, q_2 and q_3 both parallel to the x axis for $q_1 = 0$) configuration because this makes the internal torques on the joints smaller than say a ZXY configuration, as this can introduce large internal moments in the links. The ZXX configuration is equivalent to the ZYY configuration, if the angle $q_1 = \pi/2$, so there is no real reason to choose either ZXX or ZYY.

JointSensing The JointSensing block has as a main purpose obtaining the position of the end effector, velocity of the end effector and the geometric Jacobian of the end effector. It builds the geometric Jacobian on every loop iteration through an iterative approach. It works by defining the unit twists in the fixed frame, so $\hat{T}_i^{(i-1),(i-1)}$, and transforms them into the fixed frame using Brockett's equation and Rodrigues' formula. It calculates the end effector position through the joint angles, and the end effector velocity through the geometric Jacobian and the joint velocities. The only inputs to this sub-model are the joint angles, with the output given the geometric Jacobian of the end effector $J_{EE}(\mathbf{q}) \in \mathbb{R}^{6 \times 3}$, and the position vector and velocity vector of the end effector stacked into one vector $p_s = [p^T, v^T]^T \in \mathbb{R}^6$.

An argument can be made that implementing the analytical Jacobian as calculated in Equation 7 would be more numerically efficient. However, this implementation was finished before the geometric Jacobian was derived analytically. Furthermore, the system was observed to behave as expected, so it was kept like this.

Force inputs Force actuators were added to the model as signal inputs. There are a total of four, which are

1. A force in local x-direction of frame 2 on joint 2
2. A force in local y-direction of frame 2 on joint 2
3. A force in local x-direction of frame 2 on the end effector
4. A force in local y-direction of frame 2 on the end effector
5. A force in local z-direction of frame 2 on the end effector

These forces were implemented as signal lines, because it simplifies connecting a signal source to it, allowing precise control of the applied force. Furthermore, the directions and positions were chosen to test how the system behaves when a force is applied to the end effector, and to investigate what happens if a force is applied to the joint. See Figure 3 for the forces in x and y direction. The force in the z direction of frame 2 on the end effector was implemented manually using code, so it is not visible in this figure. This was done because it was implemented for quick testing, and not removed later.

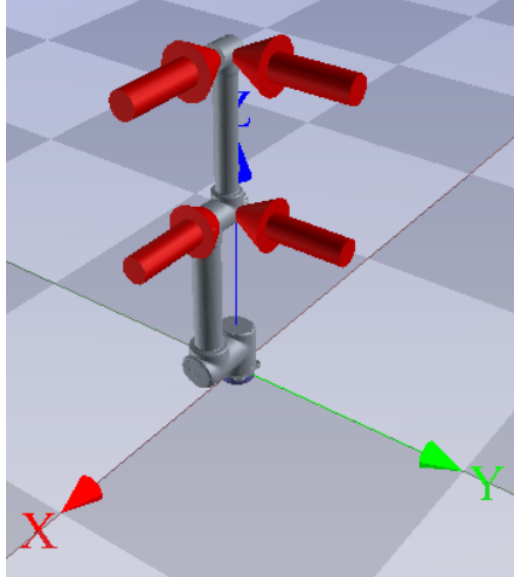


Figure 3: The forces that can be applied to the robot arm from the simulation. The forces are in the local frames attached to the third link and are displayed using red arrows.

2.3 PD Controller and Torque Calculation

2.3.1 Theoretical background

An impedance controller works like a PD controller, yielding to external forces with a spring damper behaviour. Therefore, the K_p gain can be seen as the spring constant, and the K_d gain can be seen as the damping coefficient. Using an error signal as the inputs to these gains models the system as if an external force is pulling in the end effector to the desired location. In the case of a scalar K_d and K_p , the largest pulling force is achieved if a reference changes all three coordinates (x, y, z) at the same time. Therefore, the largest output force of the end effector should be scaled with this in mind.

Next up is transforming this end effector force output of the PD controller into a torque requirement for the SEAs. To do so, the Jacobian $J_{EE}(\mathbf{q})$ is required, as derived in Equation 7 and using the following relation

$$J_{EE,linear}(\mathbf{q}) = J_{EE}(\mathbf{q})[4 : 6, 1 : 3] \quad (10)$$

$$\tau_{\mathbf{m}} = J_{EE,linear}^T(\mathbf{q})\mathbf{F}_{PD} \quad (11)$$

where $J_{EE,linear}$ is the translational part of the geometric Jacobian of the end effector, τ_m is the required motor torque and F_{PD} is the force supplied from the PD controller, which is the required force on the end effector to follow the reference trajectory.

2.3.2 Model overview

An overview of both the PD-controller block, and the torque calculation block are given in Figure 4 and Figure 5 respectively.

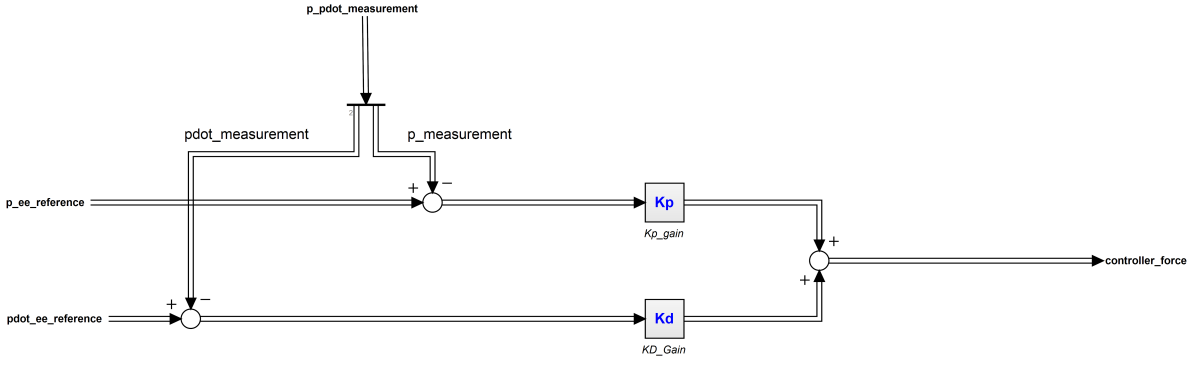


Figure 4: The PD controller (impedance controller) layout in 20-sim. The K_p and K_d gain blocks are simply a multiplication of the three states with a scalar value. The inputs to this block are the end effector reference, and velocity signals, with an extra input for current end effector position and velocity in global frame.

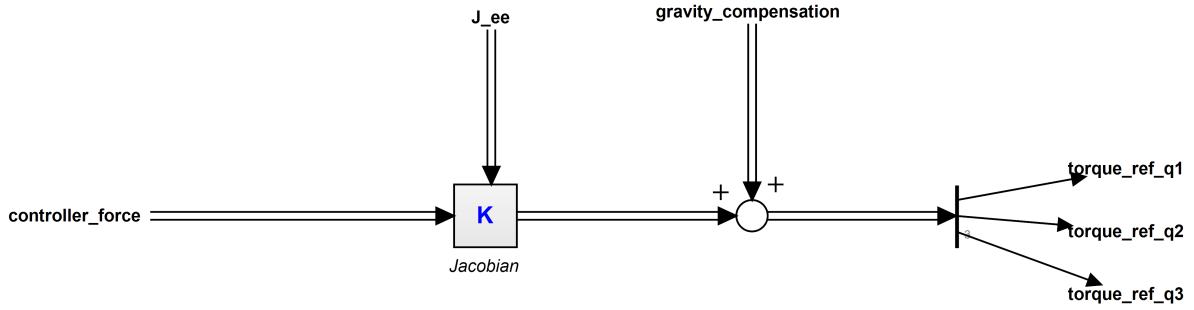


Figure 5: The torque calculation block layout in 20-sim. The Jacobian is imported, together with the required controller force, and gravity compensation. The output results in the three torques required to obtain the reference trajectory.

2.3.3 Modelling choices

The error signals are calculated using the `plusminus` blocks, which are then fed into the gains. Note that the gains are scalar values, so they respond to linear errors in all directions equally. This basically models them as if there is a spring damper combination in every direction (x, y, z) in the global frame. A K_p gain was chosen based on the stiffness requirement for the end effector, giving a final value of $K_p = 400$. To minimize oscillations of the end effector position, a damping coefficient of $K_d = 40$ was chosen, based on some trial and error. The system turned unstable if the coefficient was set lower than 40. A low value is desired for fast tracking, so the low end of the stable region of K_d was chosen. The results, and more information on the PD controller itself are given in subsection 3.2.

The Jacobian block is implemented exactly like Equation 11, so it outputs a torque reference straight to the SEAs. There is the addition of the gravity compensation term, but this will be discussed in the next subsection.

2.4 Gravity Compensation

To improve stability of the arm, and decrease steady state error of the PD controller, a gravity compensating term was added. The intent of this term is to output a motor torque for each motor on top of the PD output, to adjust for gravity along every point of the robot arm.

2.4.1 Theoretical background

Gravity can be modelled as a global frame force on each centre of mass of each link in the robot arm. To do so, a geometric Jacobian can be used, and transformed, to sum these effects.

The wrench induced by gravity on the origin of the base frame is given by

$$W = m \begin{bmatrix} \mathbf{r}_{\text{com}} \times \mathbf{g} \\ \mathbf{g} \end{bmatrix} \in \mathbb{R}^{6 \times 1} \quad (12)$$

here, $\mathbf{g} \in \mathbb{R}^3$ is the gravitational constant, with the only entry being g in the third row, and the rest just zeros. $\mathbf{r}_{\text{com}}$ is the centre of mass expressed in the fixed frame. Using this, the gravity components of all links can be calculated, and summed to get a total gravity wrench. To counteract this, one can use this wrench to transform back into the joint torques using the geometric Jacobian. In total, this yields

$$\mathbf{G}_0 = -J^T(\mathbf{q})(\mathbf{W}_{g1} + \mathbf{W}_{g2} + \mathbf{W}_{g3}) \quad (13)$$

where $\mathbf{W}_{g1}$, $\mathbf{W}_{g2}$, $\mathbf{W}_{g3}$ are the wrenches applied by body 1, 2, 3 respectively, and $\mathbf{G}_0$ is the gravity compensation term. This summation is possible due to the representation of the wrench in the world frame, which means a transformation like in Equation 7 is not necessary here.

2.4.2 Model overview

Gravity compensation is calculated in the GravityComp sub-model. It takes in the angles from the robot arm sub-model, and calculates the required motor torques to cancel out this gravity term.

2.4.3 Modelling choices

The only input to this sub-model was chosen as the joint angles. This is realistic since this is the only data that this sub-model would have access to through the rotary encoders in the joints. The implementation was done through a product of exponentials approach, just like with the JointSensing sub-model. The gravity compensation for the first joint is always set to zero, since it carries no gravitational load. For the other joints, the gravity component is calculated, and negated. This is then outputted to the rest of the model.

2.5 SEA Elements

2.5.1 Theoretical background

Every joint of the arm is driven by a series-elastic actuator (SEA): an electric motor with a gearbox and a torsion spring between the gearbox output and the joint. The deflection of the spring gives a direct measurement of the joint torque, which is what makes torque control possible. Each actuator contains two nested control loops:

$$\begin{aligned} \tau_{\text{ref}} &\rightarrow \underbrace{\text{torque controller (FF+PID)}}_{\text{spring torque}} \rightarrow i_{\text{ref}} \rightarrow \underbrace{\text{current PI}}_{\text{motor torque}} \\ &\rightarrow u \rightarrow \text{PMSM} \rightarrow \text{gearbox} \rightarrow \text{spring} \rightarrow q. \end{aligned}$$

The real UR10 uses custom-made actuators and Universal Robots does not publish their specifications, so the real parameters could not be obtained. Instead, commercial components that would physically fit inside the joints were selected and all model parameters were taken from their datasheets. For the motors, Kollmorgen KBM(S) frameless motors were chosen. The KBM-17 ($\varnothing 85$ mm) fits the base and shoulder joints, the smaller KBM-14 ($\varnothing 75$ mm) fits the elbow. For the gearboxes, Harmonic Drive CSG component sets are used [2]. Friction is the dominant loss in harmonic-drive joints and is hard to get from datasheets, so the friction model that Petrone et al. identified experimentally on a real UR10 is used [3].

2.5.2 Torque requirements and component selection

The worst case for the joints is the arm stretched out horizontally. With the UR10 parameters [1] ($m_2 = 12.7$ kg upper arm with its centre of mass at $l_{c2} = 0.232$ m, $m_3 = 4.27$ kg forearm at $l_{c3} = 0.3323$ m, link lengths $a_2 = 0.612$ m and $a_3 = 0.5723$ m, plus a 0.5 kg tool) the static gravity torques are

$$\tau_{\text{sh}} = m_2 g l_{c2} + m_3 g (a_2 + l_{c3}) + m_{ee} g (a_2 + a_3) = 74.3 \text{ Nm}, \quad (14)$$

$$\tau_{\text{el}} = m_3 g l_{c3} + m_{ee} g a_3 = 16.7 \text{ Nm}. \quad (15)$$

On top of gravity, the shoulder and elbow actuators must also overcome their own joint friction. The Coulomb friction of the friction model in Section 2.5.3 is at worst about 20 Nm for the shoulder and 10 Nm for the elbow. Adding this to the gravity torques and rounding up gives design requirements of roughly 96 Nm and 27 Nm. The

base joint rotates about a vertical axis and therefore carries no gravity torque at all. It only has to overcome friction and accelerate the arm.

Table 2 shows the selected components. The CSG-25-100 gearbox is rated for 87 Nm at 2000 rpm input, with 140 Nm average and 204 Nm repeated peak torque [2, p. 38]. The worst case of 96 Nm is above the rated torque but well below the average and peak limits. The rated torque is the torque the gearbox can deliver continuously, while the higher limits apply to torques that occur only for short times. The stretched horizontal pose is where the gravity torque is largest so in practice this means the arm can move through this pose during a motion, but it should not stay stretched out for a longer time duration because then the gearbox is loaded above its continuous rating. This is an operating limit of the design. The base gets the same actuator as the shoulder and because it is not affected by gravity torque, it automatically stays within its limits. For the elbow the CSG-17-100 (rated 31 Nm) covers the 27 Nm worst case.

Table 2: Actuator components and datasheet parameters.

	base / shoulder	elbow	source
motor	KBM(S)-17X01-C	KBM(S)-14X01-C	[4, p. 22 / p. 18]
torque constant K_t	0.354 Nm/A _{rms}	0.198 Nm/A _{rms}	[4]
resistance R (line-line)	1.54 Ω	1.26 Ω	[4]
inductance L (line-line)	5.0 mH	2.6 mH	[4]
rotor inertia J_m	$4.56 \cdot 10^{-5}$ kg m ²	$2.22 \cdot 10^{-5}$ kg m ²	[4]
static friction T_f	0.020 Nm	0.018 Nm	[4]
viscous damping F_i	$5.0 \cdot 10^{-3}$ Nm/krpm	$2.0 \cdot 10^{-3}$ Nm/krpm	[4]
continuous / peak current	6.16 / 21.4 A	6.47 / 19.85 A	[4]
gearbox ($n = 100$)	CSG-25-100	CSG-17-100	[2, p. 38]
rated / avg / peak torque	87 / 140 / 204 Nm	31 / 51 / 70 Nm	[2, p. 38]
wave-gen. inertia J_{wg}	$0.413 \cdot 10^{-4}$ kg m ²	$0.079 \cdot 10^{-4}$ kg m ²	[2, p. 38]
spring stiffness k	1600 Nm/rad	400 Nm/rad	designed
DC bus voltage U_{bus}	± 325 V (drive on 230 V _{ac})		[4]

Electrical model The KBM motors are three-phase synchronous motors, but in the model they are replaced by an equivalent DC motor: a gyrator with ratio K_t (so $\tau_m = K_t i$ and $e = K_t \omega_m$), in series with the line-to-line resistance and inductance from the datasheet. The C winding is rated for 240 V_{ac}, so it is suitable for a drive supplied from European 230 V_{ac} mains. The servo drive first rectifies the AC voltage using a diode bridge and then smooths it with a capacitor. The capacitor charges close to the peak of the mains voltage. Since 230 V_{ac} is an RMS value, this peak is

$$230\sqrt{2} \approx 325 \text{ V}.$$

The internal DC bus is therefore about 325 V. This is the voltage available to the motor drive, which is why the 240 V_{ac} C winding is suitable. Even at the UR10 joint speed limit of 120°/s, the motor speed is only 209 rad/s. This gives a back-EMF of $K_t \cdot 209 = 74$ V, which is much lower than the 325 V bus voltage. Therefore, there is enough voltage margin. The drive is modelled as an ideal voltage source limited to $\pm U_{bus}$. PWM switching is not modelled, because its frequency is much higher than the mechanical dynamics so only the average voltage is relevant.

Spring stiffness The spring stiffness was chosen by limiting the static deflection at the worst-case torque to about 0.06 rad: $k_{sh} = 96/0.06 \approx 1600$ Nm/rad and $k_{el} = 27/0.067 \approx 400$ Nm/rad. With the output locked, the motor inertia and the spring form a resonance at

$$f_{res} = \frac{1}{2\pi} \sqrt{\frac{k}{n^2(J_m + J_{wg})}}, \quad (16)$$

which is 6.8 Hz for the shoulder/base and 5.8 Hz for the elbow. This is well above the natural frequency of the Cartesian impedance loop, $f_n = \frac{1}{2\pi} \sqrt{K_p/m} \approx 1$ Hz, so the cascaded control has the frequency separation it needs to work. Because J_{wg} is a motor-side inertia, it is reflected to the gearbox output as $n^2 J_{wg}$.

2.5.3 Joint friction model

Petrone et al. [3] identified the joint friction of a real UR10 as a sigmoid function of the joint velocity:

$$\tau_f(\dot{q}) = f_o + f_v \dot{q} + \frac{f_c}{1 + e^{-\delta(\nu + \dot{q})}}. \quad (17)$$

Their parameters ([3, Table IV]) are in motor-current units, so they were multiplied by the motor drive gains K_j from [3, Table V] to obtain joint torques. The result is Table 3. Because these values were measured on the complete robot joint, they already contain the gearbox, bearing and seal losses. A separate gearbox efficiency is therefore not added. Only the motor's own small bearing friction (T_f and F_i from the KBM datasheet) is added on the motor side. A small nonzero torque remains when the robot is not moving (about -4.3 Nm for joint 2). Because it stays inside the stiction band of the joint, it is accepted as physically reasonable.

Table 3: Joint friction parameters: [3, Table IV] scaled by the drive gains K_j of [3, Table V].

joint	f_v [Nm s/rad]	f_o [Nm]	f_c [Nm]	δ [s/rad]	ν [rad/s]
1 (base)	14.85	-14.05	28.62	7.9467	-0.0185
2 (shoulder)	13.79	13.26	-33.30	-59.9536	-0.0019
3 (elbow)	7.82	-9.34	18.96	19.8251	-0.0053

2.5.4 Controller design

Current loop The electrical part of the motor is $P_i(s) = 1/(Ls + R)$. A PI controller is used, with its zero placed exactly on the plant pole ($K_{i,i}/K_{p,i} = R/L$), so the closed loop becomes a simple first-order system. The gain sets the bandwidth, which is chosen as $f_{c,i} = 500\text{ Hz}$:

$$K_{p,i} = \omega_{c,i}L, \quad K_{i,i} = \omega_{c,i}R, \quad \omega_{c,i} = 2\pi f_{c,i}. \quad (18)$$

This gives $K_{p,i} = 15.7\text{ V/A}$ and $K_{i,i} = 4838\text{ V/(A s)}$ for the shoulder motor, and 8.17 V/A and 3958 V/(A s) for the elbow. The closed current loop then has a time constant of $1/\omega_{c,i} = 0.32\text{ ms}$. Because 500 Hz is about 20 times the torque-loop crossover frequency, the torque controller effectively sees an ideal torque source. The integrator is clamped at $U_{\text{bus}}/K_{i,i}$ (anti-windup).

Torque loop With the output locked, the transfer function from motor torque to spring torque is

$$P_\tau(s) = \frac{k/n}{Js^2 + bs + k/n^2}, \quad J = J_m + J_{wg}, \quad (19)$$

with the resonance of (16). The torque controller is a feedforward plus PID:

$$\tau_m^{\text{ref}} = \frac{\tau_{\text{ref}}}{n} + K_p e + K_i \int e dt - K_d \frac{d}{dt} [\tau_s]_{\text{filt}}, \quad e = \tau_{\text{ref}} - \tau_s. \quad (20)$$

The feedforward τ_{ref}/n is the motor torque that would be needed if the actuator were ideal. The derivative term is applied to a filtered version of the measured spring torque. A first-order filter with $\tau_f = 1.27\text{ ms}$ is used to reduce noise, which becomes especially important once the torque measurement is sampled and quantised. The gains are tuned for a crossover of 25 Hz , with the lead zero at one third and the integral zero at one tenth of the crossover: $K_p = 0.0388$, $K_i = 0.609$, $K_d = 7.4 \cdot 10^{-4}$ (shoulder/base) and $K_p = 0.0549$, $K_i = 0.862$, $K_d = 1.05 \cdot 10^{-3}$ (elbow). Both actuators then have a phase margin of 61° and a closed-loop bandwidth of about 38 Hz . The integrator is clamped so that its contribution cannot exceed the peak motor torque ($K_i \cdot t_{\text{int,max}} = K_t I_p$), and the current reference is limited to the datasheet peak current I_p . This current limit is also the torque limit of the actuator.

2.6 Discretization

The entire model is now still in the continuous domain. However, in practice, parts such as the controllers function in the discrete domain. Thus, to make the model more representative of a real life robot arm, the controllers and sensors are discretized.

2.6.1 Theoretical background

For discretization, the A/D and D/A converters need a minimum and maximum value for the quantity they convert. For the motor current and the bus voltage these come directly from the datasheet (Table 2), and the joint angle uses the full $\pm\pi\text{ rad}$ range of the encoder. The only range that has to be estimated is the spring torque, which is set from the worst-case joint torque ($\approx 96\text{ Nm}$ for the shoulder). The full reasoning and the resulting converter resolutions are given in subsection 5.1.

Each control loop also needs a sample time. The sample times were chosen from the closed-loop bandwidths. A common rule is to sample at 20–30 times the bandwidth, which keeps the phase loss from the sample-and-hold

below about 10° . This results in sample times of $50\mu\text{s}$ for the current loop, 1 ms for the torque loop and 8 ms for the outer loop. The full calculation is given in subsection 5.1.

Because the controllers are discrete, the Cartesian position, velocity and gravity torque are not measured directly. They are computed inside the JointSensing and GravityComp blocks from the sampled joint angles, with the velocity obtained by differencing the quantised encoder signal. This differenced velocity carries a delay, which limits how much damping (K_d) the outer loop can use, as discussed in subsection 2.3.

2.6.2 Model overview

The overall discrete control architecture is shown in Figure 6: three nested digital loops running at different sample rates around the continuous plant.

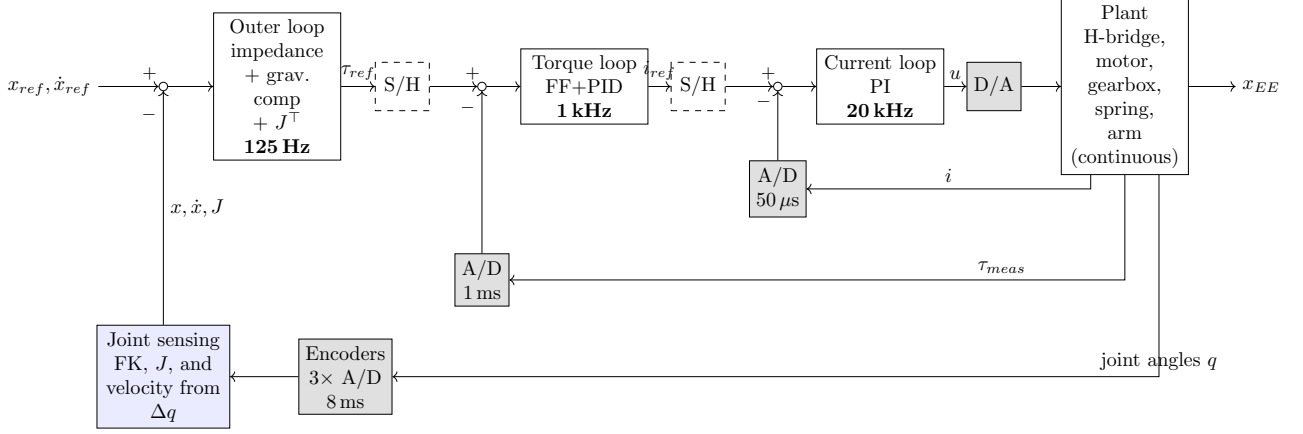


Figure 6: Discrete cascaded control architecture. The Cartesian impedance controller, including gravity compensation and the Jacobian-transpose mapping, runs at 125 Hz. The SEA torque controller runs at 1 kHz and the current controller runs at 20 kHz. The H-bridge, motor, gearbox, series spring and robot arm remain continuous. The grey blocks convert between the continuous plant and the digital controllers. The current sensor, spring-torque sensor and joint encoders sample and quantise their measurements at the rate of their control loop. The motor-voltage signal is quantised and held before it enters the continuous plant. The dashed sample-and-hold blocks pass the torque and current references to the faster loops without quantising them again. The end-effector position and velocity are calculated from the sampled joint angles in the joint-sensing block. This block uses forward kinematics, the Jacobian and a filtered velocity estimate. The delay in this velocity estimate limits the damping gain, as discussed in subsection 3.2.

The models shown below are nearly identical to the models shown in subsection 2.3 and subsection 2.5, but with the A/D and D/A converters and the Sample and ZoH blocks added for discretization.

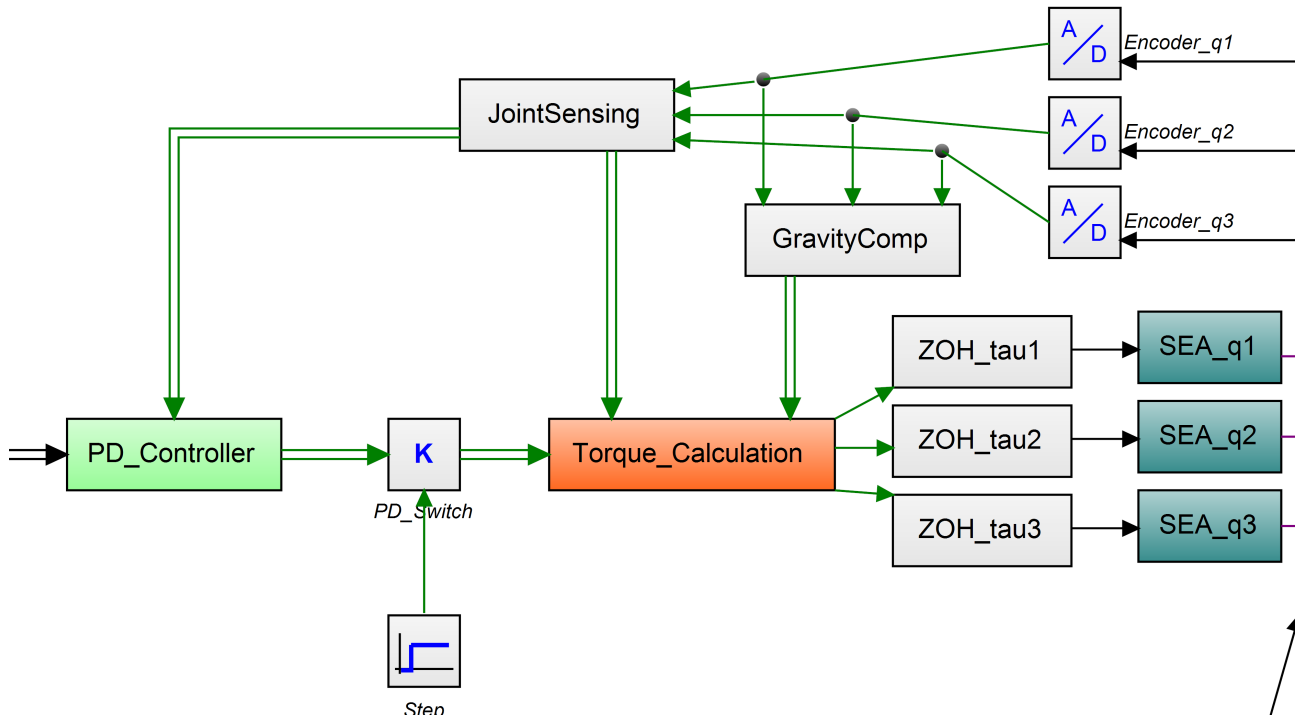


Figure 7: 20-sim structure of the discrete outer (Cartesian) control loop: the three joint encoders (A/D) feed the JointSensing and GravityComp blocks; the impedance (PD) controller and Torque_Calculation form the joint-torque references, which pass through the zero-order holds (ZOH_tau) to the three SEAs.

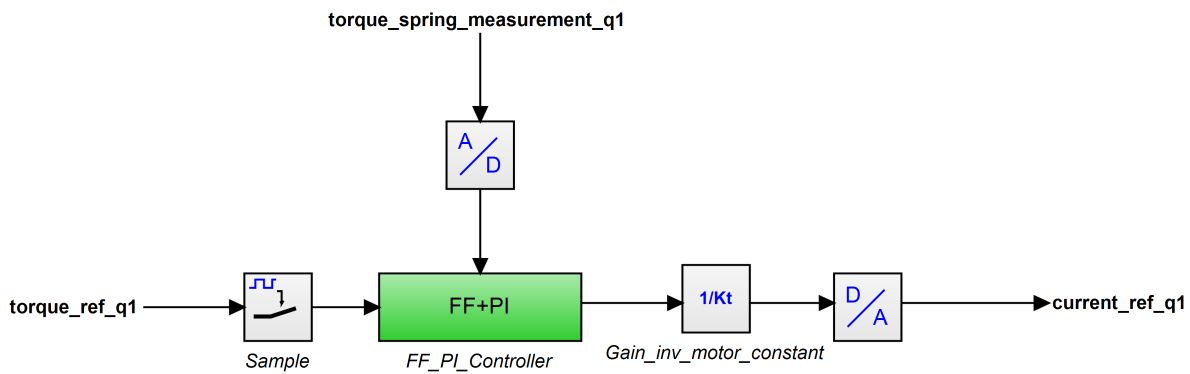


Figure 8: 20-sim model structure of the torque controller within the SEA submodel

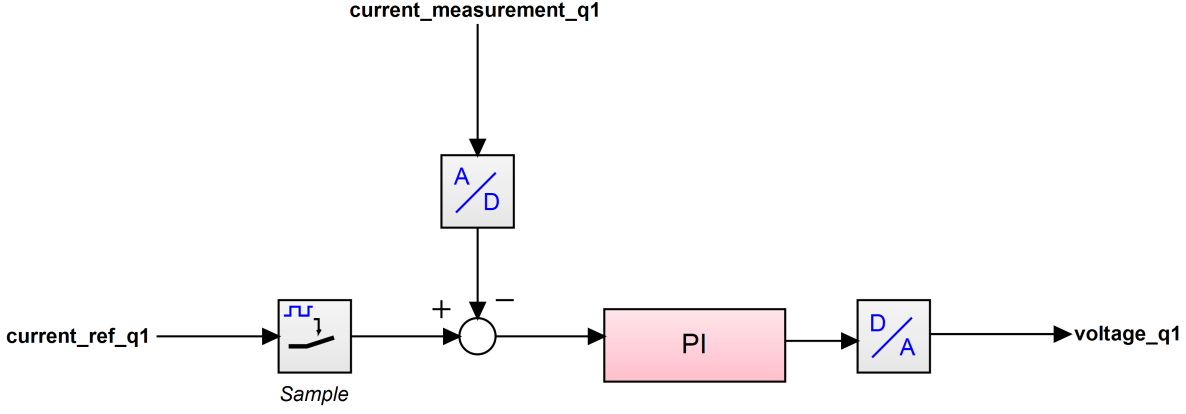


Figure 9: 20-sim model structure of the current controller within the SEA submodel

2.6.3 Modelling choices

In 20-sim, there are two simple ways to discretize a part of the model. One can either use a sample block, or an A/D converter. Sampling simply changes a continuous time signal into a discrete time signal. An A/D converter though, can represent actual hardware better as it will also quantize the signal. In quantization, the value is rounded to the nearest available number based on the amount of bits that the converter has. Signals that are not 'going through sensors' are converted with a simple sample block. A similar methodology is used for the conversion from discrete to continuous domain with D/A converters and Zero Order Hold blocks.

3 Simulation

3.1 Robot Manipulator

3.1.1 Verification

To verify that the multi-body rigid dynamics and kinematics of the robot manipulator were modeled and compiled correctly, the submodel was isolated and tested in two different experiments.

First, a free-fall experiment under the influence of gravity was done. The robot arm was released from a certain position (with an initial end-effector position of $p_{EE} = [0.40, 0.55, 0.70]^T$ with zero motor input torques. Due to the ZXX kinematic configuration of the joints, joint 2 and 3 were expected to behave like a double pendulum, while joint 1 would move a bit, but stay relatively stationary. The simulation confirmed this behavior. It showed that the link connections and joint rotation axes are mapped properly in the 3D-mechanics environment.

Second, the conservation of the total mechanical energy was verified during the same frictionless free-fall simulation. Here, it was checked if the total energy (E_{total}) would stay constant as it is the sum of the kinetic energy (E_k) and the gravitational potential energy (E_p). This was implemented in 20-sim with an equation block, with equations

$$E_k(t) = \frac{1}{2} v^\top M(q) v \quad (21)$$

$$P_g(t) = G_{\text{grav}}^\top(q) v \quad (22)$$

$$E_p(t) = \int P_g(t) dt + E_{p,\text{init}} \quad (23)$$

$$E_{\text{total}}(t) = E_k(t) + E_p(t) \quad (24)$$

The energy profiles showed a continuous exchange between potential and kinetic energy, while the total energy remained constant. The kinetic energy showed a maximum of $132.84J$, while the potential energy dropped from its initial value of $133.60J$ to a minimum of $0.02J$, while the total energy stayed around $133.60J$. This can be seen in Figure 10. The absence of energy growth confirms that the manipulator is indeed passive.

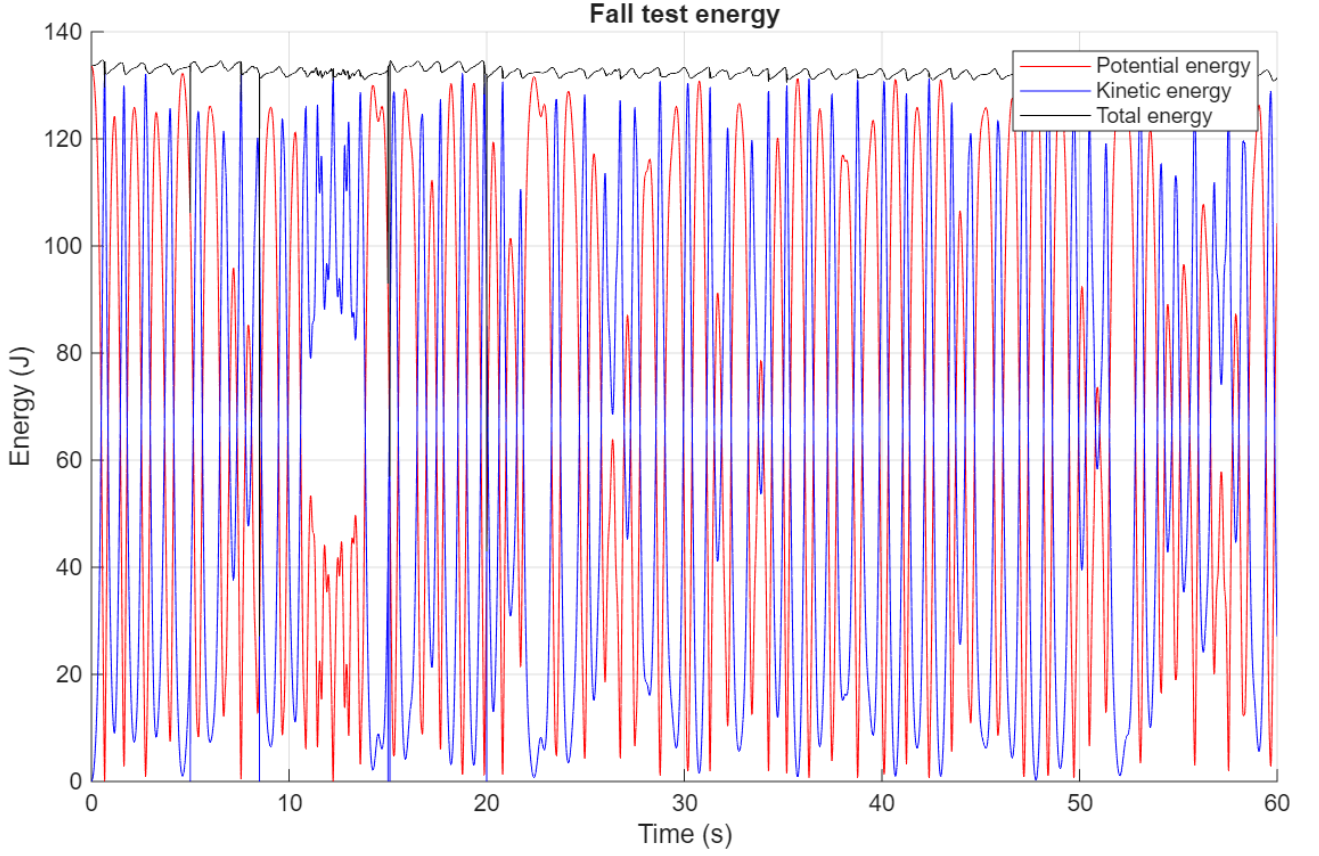


Figure 10: Total energy, kinetic energy and potential energy tracked over time when the robot arm is falling under gravity

3.1.2 Validation

Validation of the robotic arm ensures that it satisfies its operational design requirements. This was validated by evaluating the static torque requirements of the arm in its worst-case scenario, which is a completely stretched horizontal pose.

The analytical gravity torques required to maintain a static horizontal configuration were calculated in subsection 2.5 to be $\tau_{shoulder} = 74.3Nm$ and $\tau_{elbow} = 16.7Nm$. When tested in the final model, the simulation confirmed that the actuators successfully exerted the required holding torques, reaching steady-state values of $\tau_{shoulder} = 72.63Nm$ and $\tau_{elbow} = 10.74Nm$ while keeping the robot stable in the horizontal static position without sagging. This shows that physical parameters are validated for the robot's purpose, but not entirely the same as the theoretical values. This can be the result of the joints being in a stiction band of the joint when in equilibrium. The result of this experiment is shown in Figure 11.

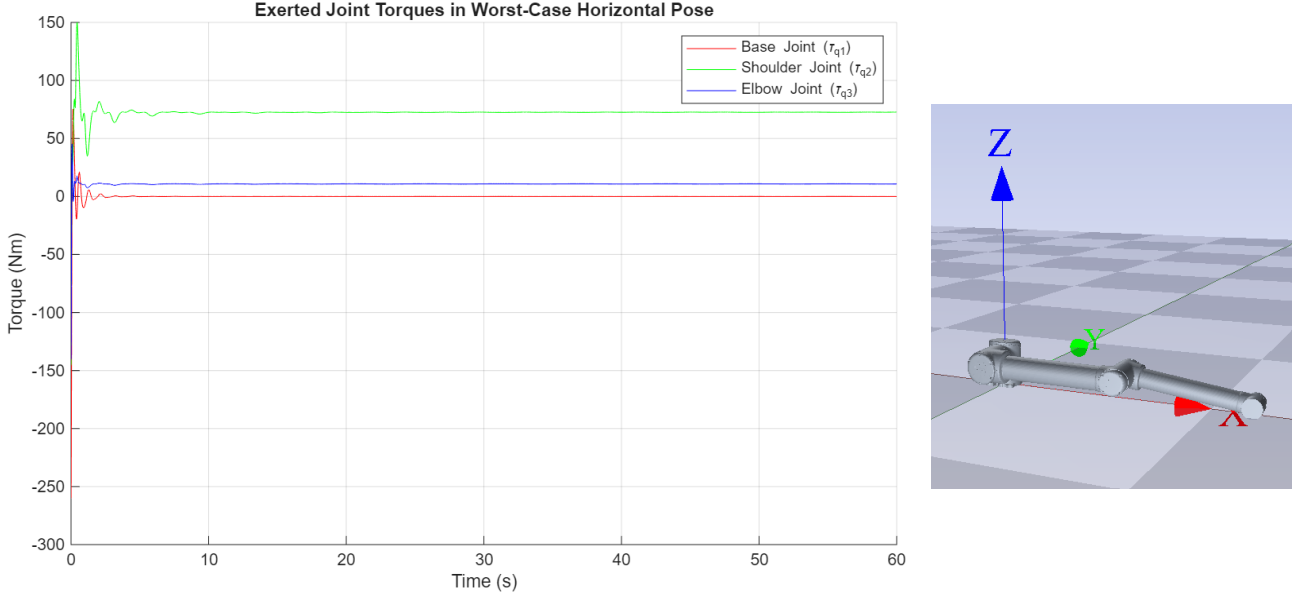


Figure 11: Robot Arm simulation analysis showing torques of the joint in a stretch out position.

3.2 PD Controller and Torque Calculation

Two properties have to be checked for this block: that the impedance law produces the correct restoring force from a position error and that this force is mapped to the right joint torques through the Jacobian transpose of Equation 11. After that, the controller is validated on the actual 3-D tracking task and on an environment-interaction experiment.

3.2.1 Verification

Spring law and force-to-torque mapping The controller was tested by first letting the arm settle at its initial pose and then applying a small step of $\Delta x = 0.02$ m to the desired x -position. Right after the step the end-effector has not moved yet, so the position error is known, and the velocity is still close to zero, which means the damping term has almost no effect. The Cartesian force could therefore be compared with the force expected from the stiffness and the resulting joint torques with those from the Jacobian transpose of Equation 11. To remove the gravity-compensation term, the change in each signal from just before to just after the step was used.

The controller force jumped by 8.00 N in x (and ≈ 0 in y, z), which matches $K_p \Delta x = 400 \times 0.02 = 8.0$ N to within 0.1% and confirms the impedance stiffness. The torque references jumped by $[-4.40, -2.42, 0.13]$ Nm. These values match $J_{EE,linear}^T(\mathbf{q}) \Delta \mathbf{F}$, calculated offline in MATLAB at the step configuration $\mathbf{q} = (-0.555, -0.175, -1.448)$ rad. The difference between the calculated and simulated torques was below 10^{-4} Nm. Both checks are summarised in Table 4 and together they verify that the impedance law and the Jacobian-transpose mapping are implemented correctly.

Table 4: Step-test verification of the controller: predicted vs. simulated jump of each signal across a 0.02 m position step.

check	quantity	predicted	simulated
stiffness ($F = K_p \Delta x$)	force jump ΔF_x	8.00 N	8.00 N
mapping ($\tau = J^T F$)	$\Delta \tau_{q1}$	-4.396 Nm	-4.396 Nm
	$\Delta \tau_{q2}$	-2.416 Nm	-2.416 Nm
	$\Delta \tau_{q3}$	0.127 Nm	0.127 Nm

Damping (velocity term) A 10 N force was applied to the end effector for 2 s while it was held at a fixed pose. After the force was removed, the end effector moved back towards its reference position. With $K_d = 40$, it settled without continued oscillations. For the stiff setting, this took about 0.2 s. This shows that the velocity term damps the motion. The deflection and recovery are shown in Figure 13.

3.2.2 Validation

The purpose of the controller is to make the end-effector follow a 3-D reference while staying compliant towards its environment. Both were tested on the full model with the discrete control loops active.

The model was run for two full laps (12.57 s) on the circular reference, with centre (0.20, 0.55, 0.70) m and radius 0.20 m. After a brief startup transient of about 1.5 s, the end effector tracks the circle with a mean position error of about 5 mm. During this transient, the arm accelerates from rest while the reference is already moving at 0.2 m/s. There was no visible drift over the run. The error trend was about +0.03 mm/s, and the paths from both laps overlay almost exactly. The largest error of about 17 mm occurs at the same point on each lap, at the point closest to the robot base. At this point, the effective Cartesian inertia is highest ($m_x \approx 11$ kg), so the loop is least damped. This is discussed further in the tuning section below. Figure 12 overlays the measured path on the reference. The motor currents stay well below the datasheet peak currents: 0.7 A, 2.7 A and 1.6 A, compared with 21.4 A, 21.4 A and 19.85 A. The highest current is therefore below 14% of its peak value, so the task is achieved well within the actuator limits.

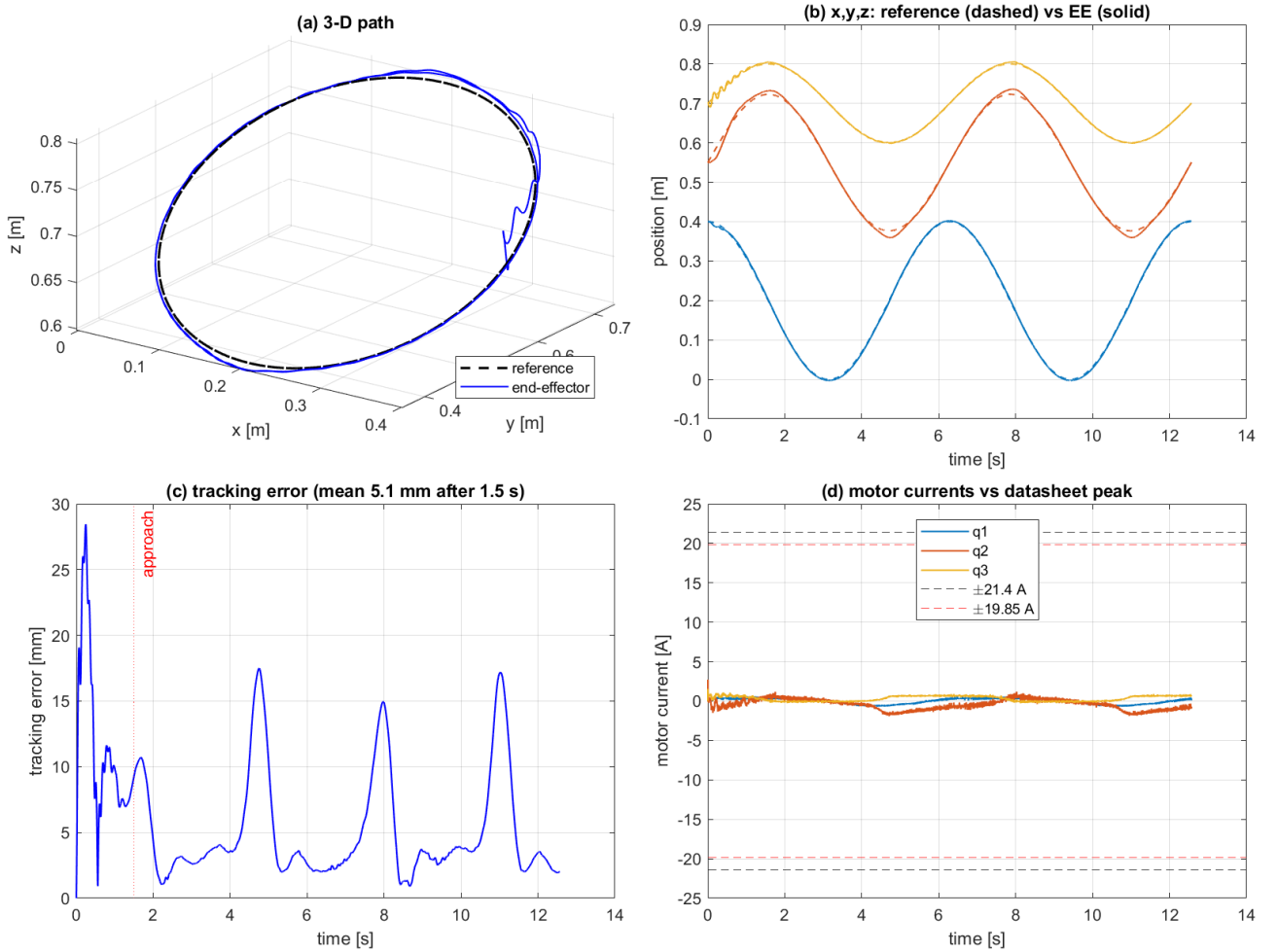


Figure 12: Circle-tracking validation (two laps, $K_p = 400$, $K_d = 40$): (a) measured end-effector path vs. reference, (b) x, y, z over time, (c) tracking error, (d) motor currents against the datasheet peaks. Mean error ≈ 5 mm, currents below 14% of the limits.

Environment interaction: stiff vs. compliant The same 10 N force pulse was applied twice while the arm was held at its initial pose. One run used a stiff setting with $K_p = 400$ and the other used a compliant setting with $K_p = 100$. In both cases, $K_d = 40$. With the stiff controller, the end effector moved about 25 mm. With the compliant controller, it moved about 99 mm. This is about four times larger, which matches the difference in stiffness. In both cases, the final deflection was close to F/K_p , as expected for a Cartesian spring. After the force was removed, both controllers moved the end effector back to the reference position, as shown in Figure 13. The compliant controller returned more smoothly because lowering K_p while keeping K_d constant increases the damping ratio from about 0.3 to 0.6.

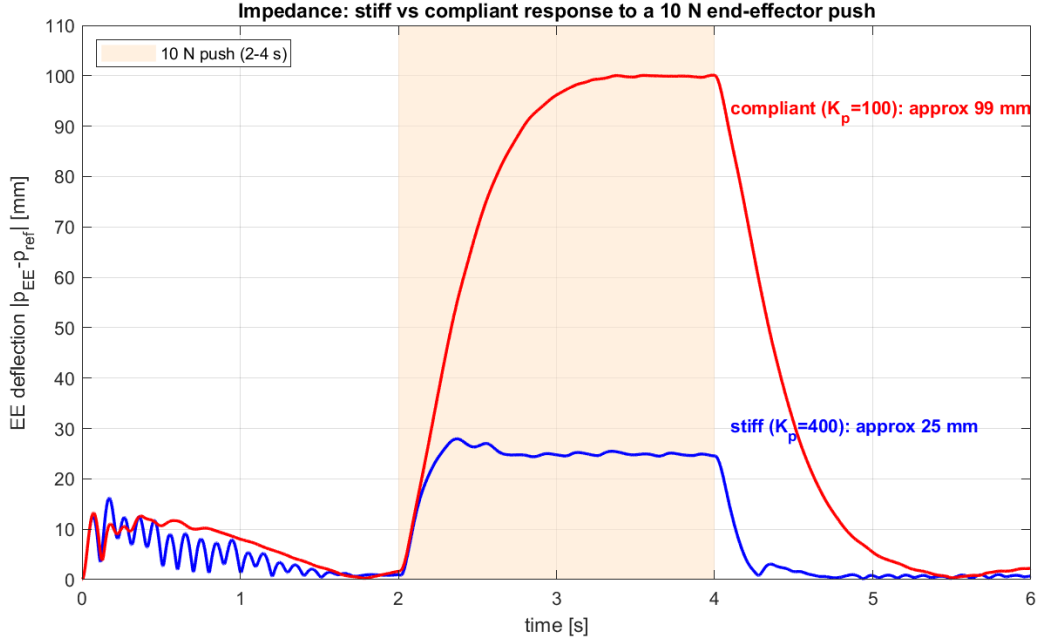


Figure 13: End-effector deflection during a 10 N force pulse from 2 s to 4 s. The stiff setting ($K_p = 400$) moves about 25 mm, while the compliant setting ($K_p = 100$) moves about 99 mm. Both return to the reference position after the force is removed.

Tuning of K_d The damping gain could not be increased without limit. With the discrete control loops, $K_d = 40$ was stable. At $K_d = 45$, the tracking error slowly increased by about $+0.06$ mm/s. At $K_d = 60$, the model became unstable. The growing oscillation had a frequency of about 6 Hz. This is close to the SEA elastic resonance of 5.8 Hz shown in Table 5.

The main cause is the velocity estimate. In the discrete model, the end-effector velocity is calculated from the sampled encoder signal using a difference and a low-pass filter. This adds a delay to the velocity signal. As a result, a larger K_d no longer reduces the SEA resonance, but instead makes it larger. For this model, $K_d = 40$ was the highest stable damping gain. This also explains the small remaining oscillation in the tracking result. The limit comes from the discrete controller, since it does not appear in the continuous version of the model, as discussed in the discretisation section.

3.3 Gravity Compensation

3.3.1 Verification

To verify the modelled gravity compensation, two distinct tests were executed.

The first verification test evaluated if the arm was able to stay in a static position. The manipulator was driven to an arbitrary position in the workspace with the PD controller, which was $[0, 0.6, 0.6]$. After reaching the setpoint, the PD controller was turned off, leaving the system under only the influence of τ_{grav} . When conditions are ideal, perfect gravity compensation would yield a net joint acceleration of zero. Over a simulation duration of 7000 s, the joints slowly drifted. The end-effector stayed in position for about 66 s. The lack of sudden dropping or rapid acceleration validates that the computed feedforward gravity torques are effective in cancelling out the gravity forces on the link masses.

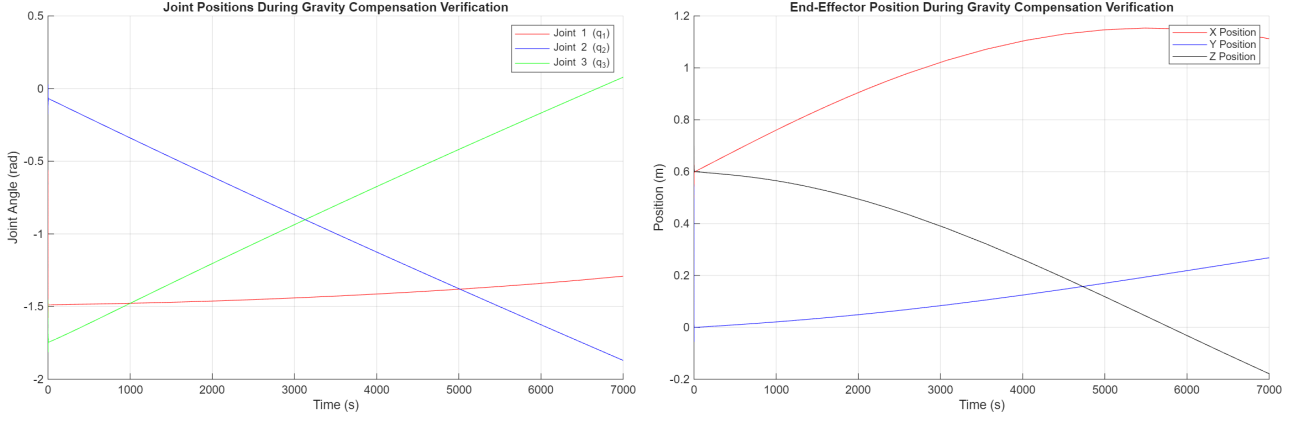


Figure 14: Gravity compensation verification results showing a) joint drift and b) end-effector position drift over time.

The second verification test assessed the dynamic compliance of the system when it is only gravity compensated. Again, after reaching the desired setpoint in the workspace, the PD controller was turned off. Then a force impulse of $10N$ was applied along the x-direction of the end-effector. The trajectory that resulted showed a coupled rotational movement and a gradual descent along the vertical z-axis. This is due to the momentum that is built up in the robot arm from this sideways force application. Due to the linear momentum induced by the force, the robot arm experiences centrifugal forces, pushing the end effector outwards. This, in turn, induces a downward momentum in the links connected to joint 2, which results in a complex coupled trajectory shown in Figure 15.

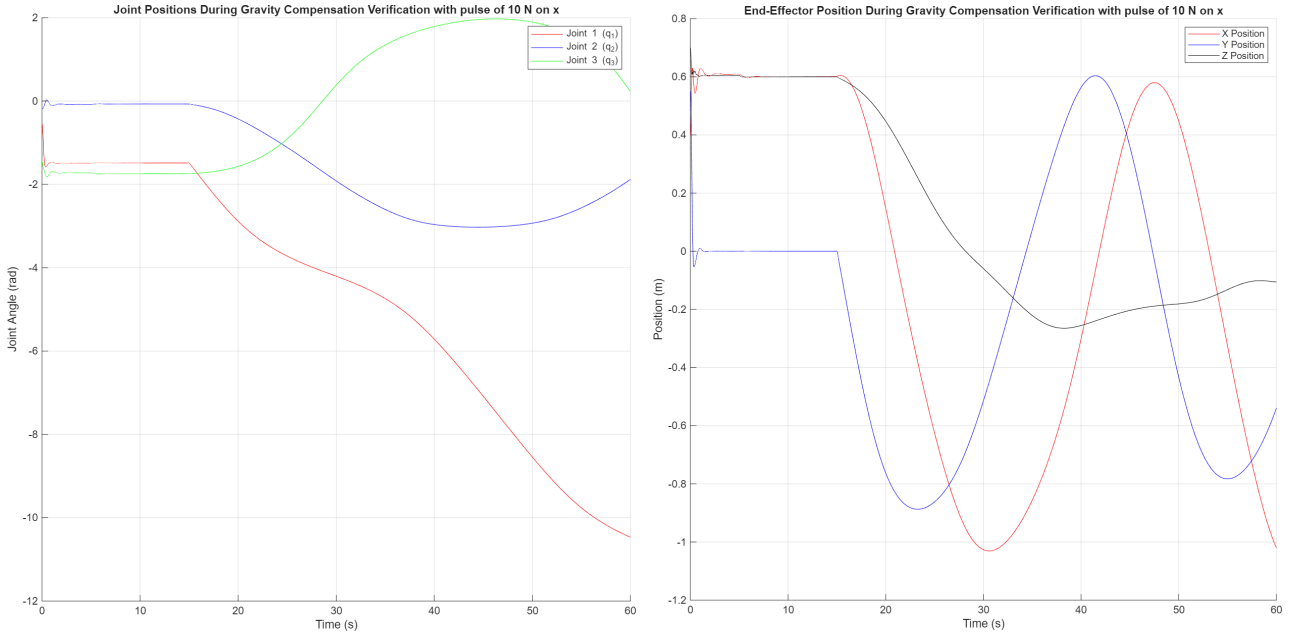


Figure 15: Gravity compensation verification results with an impulse of 10 Newton on the X of the end effector showing a) joints of the robotic manipulator over time b) end-effector position change over time.

It can be seen that the arm behaves as a passive, dissipative system that does not accelerate under its own weight, so the gravity compensation is verified.

3.3.2 Validation

To validate the operational purpose of the gravity compensation loop, namely, the elimination of steady-state configuration errors under long gravitational loading, closed-loop step response experiments were executed. The total control effort that is applied to the actuators is defined by:

$$\tau = \tau_{feedback} + \tau_{grav} \quad (25)$$

When executing a step response with the gravity compensation deactivated, the system showed a noticeable steady-state tracking error of $63.94mm$, with the end-effector settling at a position of $p_{ee} = [0.6282, 0.0000, 0.5426]^T m$ instead of the desired target of $[0.6, 0.0, 0.6]^T m$. Since the PD controller does not have the integral term needed to reject the gravity disturbance torques, the links settle of their demanded positions.

Conversely, when the gravity compensation is active, the gravitational potential field is decoupled from the feedback dynamics. This reduces the steady-state tracking error to $0.18mm$, with a final end-effector position of $p_{ee} = [0.6001, 0.0000, 0.6002]^T$, which validates that the subsystem fulfils the performance criteria. The figure of these results that shows the final positions of the end-effector with and without gravity compensation is Figure 16.

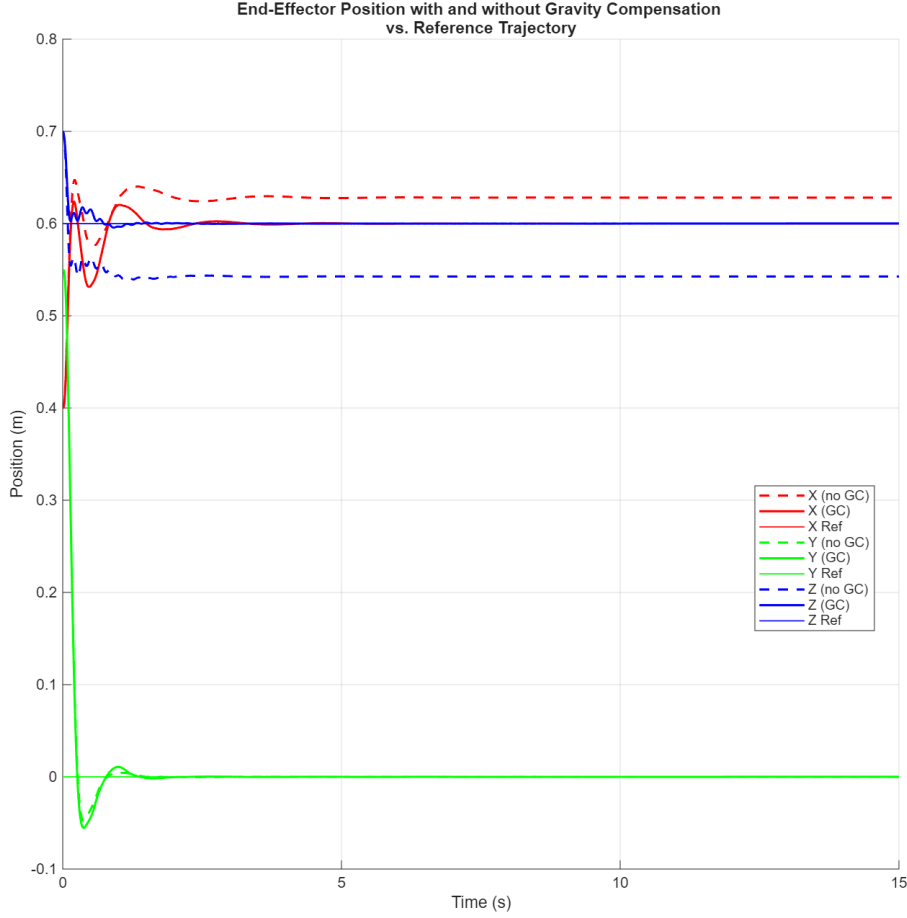


Figure 16: End-effector position tracking during a step response, comparing the performance with active gravity compensation (solid lines) and deactivated gravity compensation (dashed lines) against the reference trajectory (thin lines)

3.4 SEA Elements

3.4.1 Verification experiments

Each actuator was tested separately in an experimental setup: a torque step source, the actuator, a load inertia, the joint friction of Equation 17 and an external torque source. No arm and no outer control loop is involved. We did four experiments and for each one we first estimated the expected values. Table 5 summarises predictions and simulation results.

Table 5: Predicted vs. simulated verification results.

experiment	quantity	predicted	simulated
current step	time constant	0.32 ms	0.32 ms
torque step, free load	terminal velocity (sh)	1.214 rad/s	1.214 rad/s
	terminal velocity (base)	0.434 rad/s	0.434 rad/s
	terminal velocity (el)	0.159 rad/s	0.159 rad/s
	steady spring torque	$= \tau_{\text{ref}}$	exact
locked, closed loop	overshoot (sh / base / el)	26.4 / 26.4 / 25.3 %	23.5 / 22.9 / 23.7 %
locked, open loop	resonance (sh / base / el)	6.8 / 6.8 / 5.8 Hz	5.8 / 5.8 / 4.7 Hz
	Coulomb decay (sh)	8 Nm/cycle	≈ 10 Nm/cycle
	stick offset (sh / base / el)	2 / 2 / 1.8 Nm	1.9 / 2.0 / 1.7 Nm

Current loop (Figure 17) A step on the torque reference produces a step on the current reference. The measured current follows it as a first-order response with the designed 0.32 ms time constant and no overshoot. This confirms the electrical parameters (L/R pole cancelled by the PI zero) and shows that the currents are realistic. The largest current in any experiment was 3.9 A so far below the 6.2 A continuous rating of the motor.

Torque step against a free load (Figure 18) With the torque loop closed, a constant torque reference accelerates the load until the joint friction balances the applied torque. The final velocity follows from solving $\tau_f(\omega_{ss}) = \tau_{\text{ref}}$. For the shoulder with 30 Nm this gives $\omega_{ss} = (30 - 13.26)/13.79 = 1.214$ rad/s. All three tests match their prediction. The experiment verifies the torque controller, the gear ratio and the friction implementation. The spring deflection stays below its 0.06 rad design limit and the current stays also well below its limit. So the actuator stays within all operating limits.

Locked output, closed loop (Figure 19). With the output locked, the torque loop is excited at its most resonant condition. The simulated step response (23–24 % overshoot, settled within about 0.05 s) is very close to the step response of the linear design model (25–26 % overshoot), which is shown in the figure. The small difference is caused by motor stiction, which adds some damping that is not included in the linear model.

The 6.8 Hz resonance is not visible in the closed-loop response. The spring torque has one overshoot and then settles, instead of continuing to oscillate at the resonance frequency. This shows that the derivative term adds enough damping to suppress the resonance.

Locked output, open loop (Figure 20). To show the physical resonance more clearly, the locked-output experiment with the PID switched off were repeated. Only the feedforward term was left active. In this case, the spring torque clearly starts to oscillate.

The response matches what is expected from dry friction. First, the oscillation amplitude decreases almost linearly, by about $4T_{fn} \approx 8$ Nm per cycle. The measured peaks were 54, 43, and 34 Nm. With viscous damping, the amplitude would decrease exponentially instead. Second, the motion stops slightly below the reference torque. The final value was 28.1 Nm, while the reference was 30 Nm. This difference is close to the expected stiction torque of $T_{fn} \approx 2$ Nm. For the elbow, friction removes about 7.2 Nm of oscillation per cycle. Since the applied torque step is only 10 Nm, most of the oscillation is lost after one cycle. So the elbow joint motion stops almost immediately.

current loop: measured vs first-order model ($\tau = 0.32$ ms)

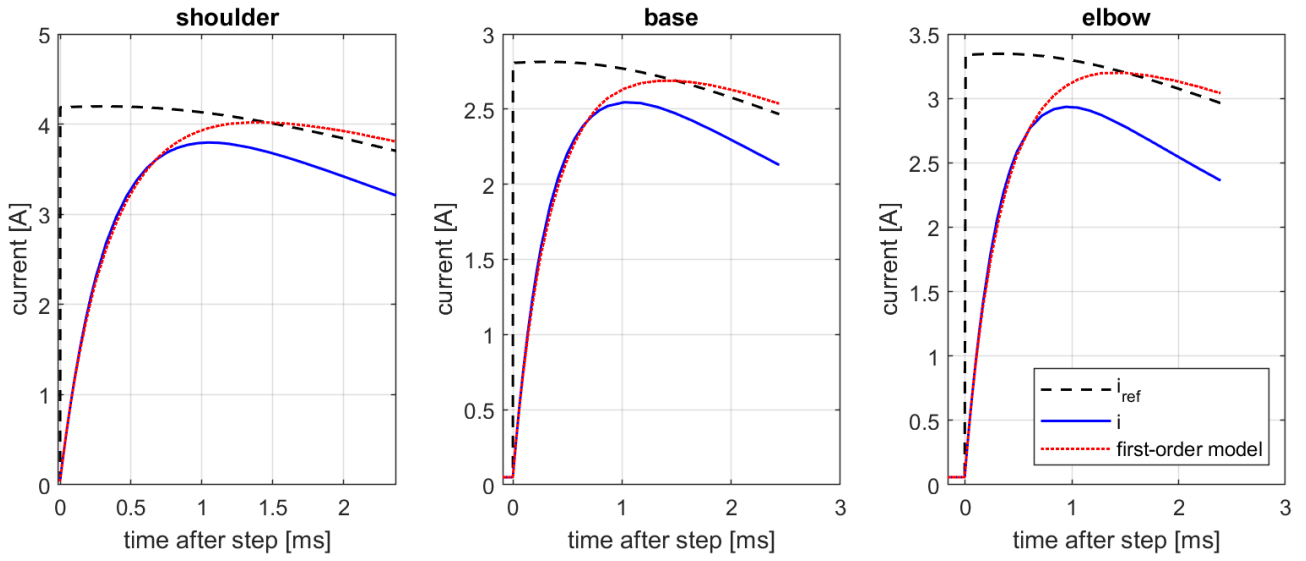


Figure 17: Current loop verification: measured current vs. the first-order model with $\tau = 0.32$ ms for the three test benches.

shoulder actuator: 30 Nm torque step, free load

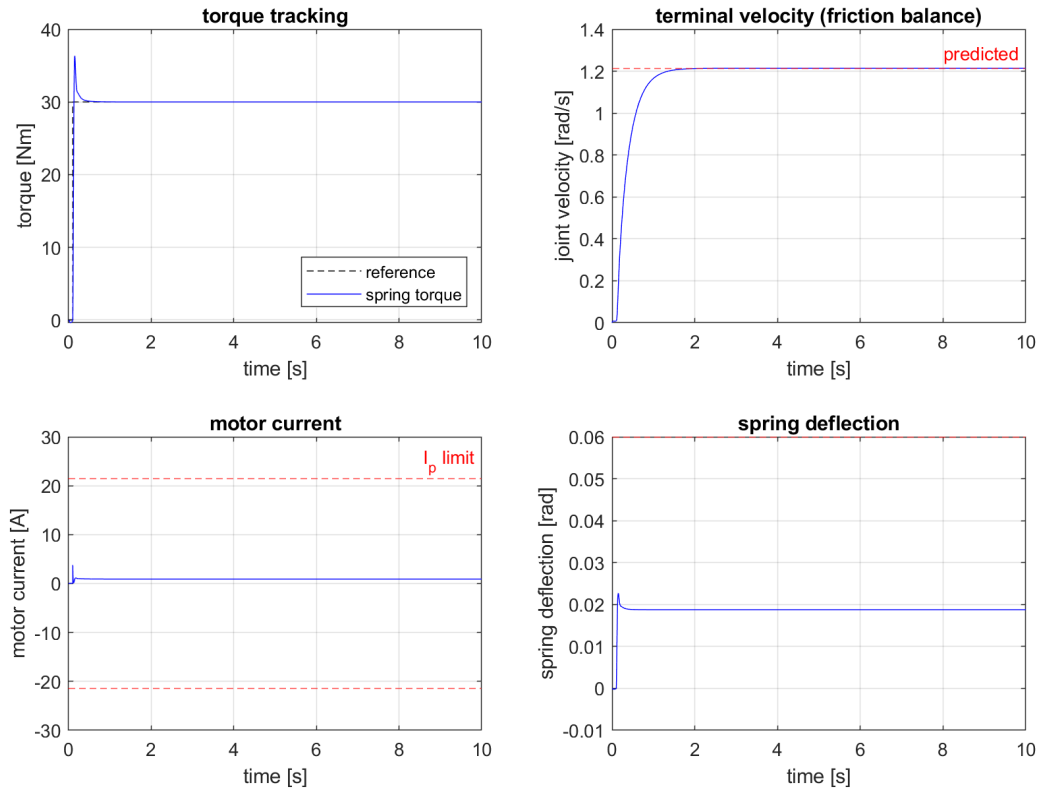


Figure 18: Shoulder actuator, 30 Nm torque step against a free load: torque tracking, terminal velocity vs. friction-balance prediction, motor current vs. peak limit, and spring deflection vs. design bound.

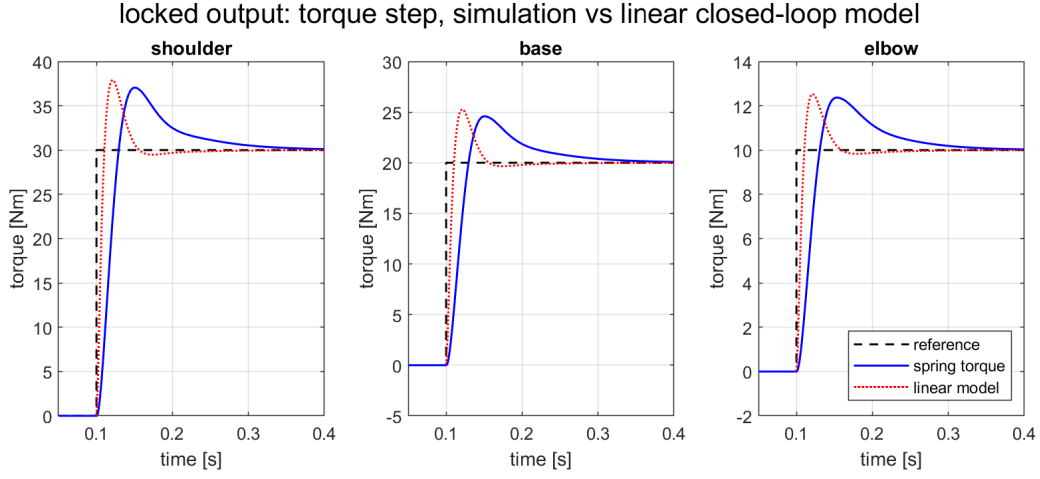


Figure 19: Locked output, closed torque loop: simulation vs. linear closed-loop model.

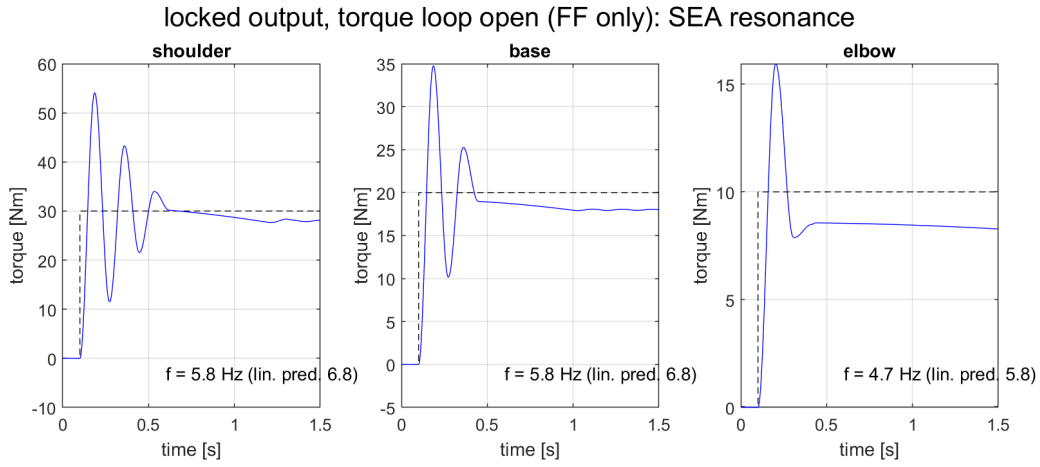


Figure 20: Locked output with the torque loop open (feedforward only): the SEA resonance can be seen, with linear (Coulomb) amplitude decay and the stiction offset

3.5 Discretization

Since the continuous model has been properly verified, it would only be useful to do the validation of the discretization to ensure that this has been implemented properly.

3.5.1 Validation

To validate the discretization, the end-effector position and velocity of the discrete model are compared with those of the continuous model to check if the model still has the same performance. This is shown in Figure 21 and Figure 22.

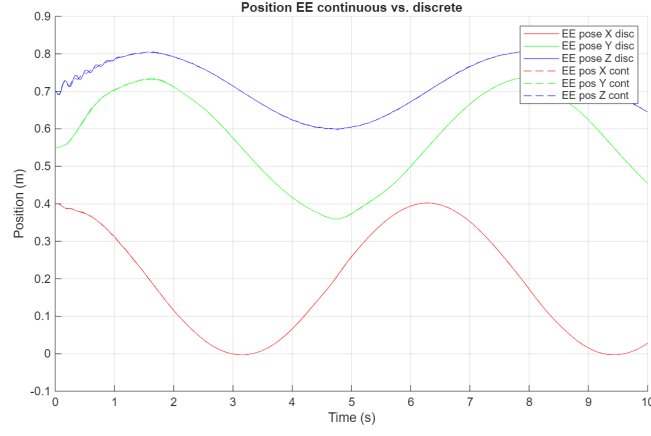


Figure 21: Plots showing the end-effector position both the fully continuous model and the model with discrete implementations

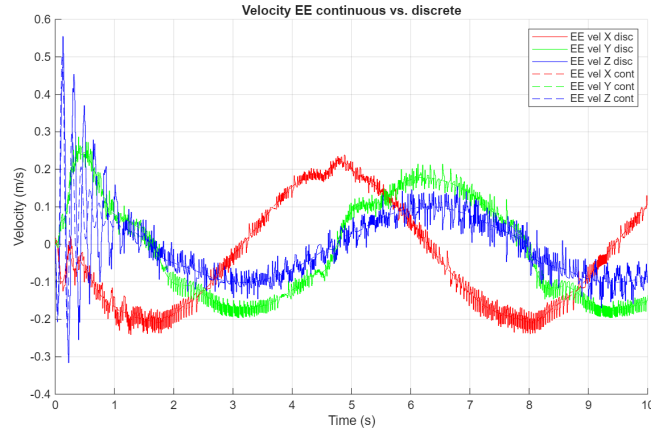


Figure 22: Plots showing the end-effector velocity of both the fully continuous model and the model with discrete implementations

The torque output of a SEAs is also checked against the continuous model. This is shown in Figure 23

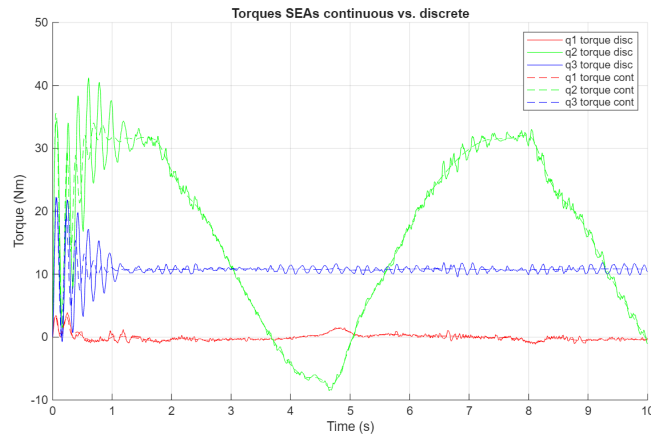


Figure 23: Plots showing the end-effector position and velocity of both the fully continuous model and the model with discrete implementations

Figure 21 shows that compared to the continuous model, the discrete has very little changes as shown to be nearly identical except for a small oscillation in the z-axis of the end-effector. Figure 22 shows that the velocity of the discretized model follows the continuous model decently well, due to the discretization, the velocity is in the shape of very small pulses. It also shows that for the first second, there is some deviation in the speed in the z-direction. Figure 23 shows that the torques of the actuators located at the upper arm and forearm do not

match that of the continuous model exactly, explaining the deviation in the start in the z-axis. After this first second, the discretized model settles and follows the same trend and values as the continuous model.

4 Discussion & Conclusion

The following conclusions can be drawn:

- The robotarm is correctly modelled with realistic parameters for the application
- The PD Controller is correctly modelled in a realistic way. It shows good performance for the application and can track the trajectory accurately, though after discretization, it is not as robust, thus it could be tuned better or better actuators would have to be chosen
- The gravity compensation is modelled in a simple and perhaps not realistic way. It works really well, but for future modelling it could be improved to resemble reality better.
- The SEA has been researched in depth to be as realistic as possible. The verification done shows sufficient accuracy, though a few more tests could be done, where a single SEA is tested with a fixed load and fixed output to test the transparency.
- The discretization is implemented correctly and the results obtained are realistic, there is a slight difference with the continuous model in the start, but this within acceptable margins.

Overall, modelling this system step by step and verifying and validating it after each stage proved to be a useful method for modelling this system. Modelling discrete systems gave a good feel on the difficulty with transforming from continuous to discrete systems, and the problems that could introduce. Furthermore, integrating separate systems together proved to bring a lot of challenges, and integration issues that were helpful in better understanding how to build models in a way that they can be fit together nicely when integration must happen later down the line. The implementation of impedance control was helpful in understanding the controller behaviour as a spring damper system, which gives such a controller a much more intuitive feeling. Apart from that, using SEAs as torque sensors is something that will be useful for future. Overall, implementing this 3DOF robot arm has improved the teams overall understanding of modelling, and the challenges that come with it.

5 Appendix

5.1 Discretisation: sample rates and converter ranges

This appendix explains how the discrete settings used in section 2 were chosen: the sample time of each control loop and the range of each A/D and D/A converter. The control structure is the one shown in Figure 6. The same reasoning is used for every loop: first estimate how fast the loop is (its closed-loop bandwidth) and then sample fast enough that the delay introduced by sampling does not noticeably reduce its stability margin.

Loop bandwidths The closed-loop bandwidth f_{BW} is the highest frequency at which a loop can still follow its input, and it is the value that decides how fast the loop has to be sampled.

The current loop sees the motor as a resistance and inductance in series, $P_i(s) = 1/(Ls + R)$, which has a single pole at $s = -R/L$. The PI controller is tuned so that its zero cancels this pole, which leaves a simple first-order loop whose bandwidth equals the chosen crossover frequency of 500 Hz. The torque loop acts on the series spring: with the output locked, the motor inertia and the spring form a resonance near 6–7 Hz, and the FF+PID controller is tuned for a crossover around 25 Hz, giving a closed-loop bandwidth of about 38 Hz. The outer loop is the Cartesian impedance controller. It makes the tip behave like a mass connected to a spring and damper. With $K_p = 400$, $K_d = 40$ and an effective tip mass of about 10 kg, its natural frequency is $\omega_n = \sqrt{K_p/m} \approx 6.3 \text{ rad/s}$. This is roughly 1 Hz.

The three loops have bandwidths of 500, Hz, 38, Hz and 1, Hz. This means that each inner loop is much faster than the loop around it, which makes sure that the cascaded controller structure works properly.

Sample rate The sample rate must be high enough compared to the bandwidth of the control loop. Between two samples, the zero-order hold keeps the control signal constant. This acts like a delay of about half a sample period. At the loop bandwidth, this delay causes a phase lag of

$$\varphi_{\text{lag}} = \pi \frac{f_{BW}}{f_s}, \quad (26)$$

so only the ratio between the sample rate and the bandwidth matters. If the sample rate is only ten times the bandwidth, the phase lag is about 18° . Sampling at twenty to thirty times the bandwidth keeps this below about 10° . Therefore, the sample rates in this model were chosen to be at least twenty times the bandwidth of their control loop.

Chosen rates Multiplying each bandwidth by twenty to thirty and rounding up to a practical value gives the rates in Table 6.

Table 6: Sample-rate selection for the three loops.

loop	f_{BW}	chosen f_s	$T_s = 1/f_s$	ZOH lag at f_{BW}
current	500 Hz	20 kHz	$50 \mu s$	4.5°
torque	38 Hz	1 kHz	1 ms	6.8°
outer	1 Hz	125 Hz	8 ms	1.4°

The current and torque loops use rates close to the minimum values from the bandwidth rule. Increasing these rates further would mainly increase the simulation time without changing the result. The outer loop runs at 125 Hz, which is much faster than required by its bandwidth. This rate is easy to simulate and gives a better velocity estimate from the sampled encoder angles. It also matches the control rate used by the UR10.

The rates were chosen as integer multiples of each other. The 125 Hz outer loop is multiplied by 8 to give the 1 kHz torque loop. The 1 kHz torque loop is multiplied by 20 to give the 20 kHz current loop. Because the rates are integer multiples, the updates of the three loops occur at the same times every 8 ms. This makes the timing between the loops easier to implement and keeps the model with different loop rates simple.

Converter ranges and resolution Each A/D converter samples a signal and rounds it to a fixed number of bits. Values outside the chosen range are clipped. The step size of the converter is

$$\text{LSB} = \frac{\max - \min}{2^{\text{bits}} - 1}. \quad (27)$$

A D/A converter converts a digital value back into a continuous signal. All converters in the model use 12 bits. The ranges were chosen to cover the expected signal values without making the range unnecessarily large, since a larger range also gives a larger step size.

The joint-angle encoder uses a range of $\pm\pi$ rad. The Cartesian position, velocity and gravity torque are not measured directly. They are calculated from the sampled joint angles in the JointSensing and GravityComp blocks. The velocity is calculated from the difference between two encoder measurements. This is also the main reason why the damping gain is limited, as discussed in subsection 3.2.

The spring-torque ranges were chosen from the expected joint torques. The largest torque occurs at the shoulder and is about 96 Nm. The base and elbow therefore use a range of ± 100 Nm. The shoulder uses a larger range of ± 200 Nm to prevent clipping during transient peaks. During the circular trajectory, the shoulder torque reaches about 40 Nm.

The motor-current range is based on the peak current from the motor datasheet. The base and shoulder use ± 21.4 A, while the elbow uses ± 19.85 A. The voltage command uses the DC bus range of ± 325 V. The resulting converter ranges and step sizes are listed in Table 7.

Table 7: Settings entered in each discrete (A/D, D/A) block and the resulting quantisation step.

block (signal)	range [min, max]	bits	T_s	LSB
A/D joint angle (encoder, outer)	$\pm\pi$ rad	12	8 ms	1.53 mrad
A/D spring torque (base, elbow)	± 100 Nm	12	1 ms	49 mNm
A/D spring torque (shoulder)	± 200 Nm	12	1 ms	98 mNm
A/D motor current (current loop)	± 21.4 A	12	$50 \mu s$	10.5 mA
D/A current reference (torque $\rightarrow$ current)	± 21.4 A	12	$50 \mu s$	10.5 mA
D/A voltage command (controller $\rightarrow$ H-bridge)	± 325 V	12	$50 \mu s$	0.16 V

The velocity is found by dividing the change in angle by the sample time. Since the encoder can only resolve steps of 1.53 mrad, over the 8 ms outer period the velocity estimate changes in steps of about 0.19 rad/s. Together with the delay added by the velocity filter, this is the reason the damping gain K_d cannot be made very large.

References

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